GROUP ASSIGNMENT

TECHNOLOGY PARK MALAYSIA

CT119-3-2- DMPM

DATA MINING AND PREDICTIVE MODELLING

GROUP L

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1.0 Introduction

Today with the advent of big data business generates a lot of data everyday out of its operations, customer interaction, and online activities. The ability to derive actionable insights from the data has become a crucial requirement to make better decisions, increase operational efficiency, and gain competitive edge. One of the key tools in this process is data mining, which uses analytical tools to find patterns, trends, and relationships in large volumes of data.

In this project, we are going to predict the student dropout risk with the STUDENT_DROPOUT_FINAL data that contains 4424 data points and 39 variables. This is a binary classification problem since the target variable DropoutRisk is binary, meaning it indicates whether a student will drop out of school or not. The project is an analysis of the factors associated with students and the construction of models using predictive modelling methods in the SAS Viya Model Studio which can identify students at risk. The findings will provide education providers with lessons for them to learn ahead of time to help minimize the proportion of pupils dropping out of school and improve pupil attainment.

1.1 Dataset Overview

The STUDENT_DROPOUT_FINAL dataset is the one that was created and uploaded to Learners on SAS Viya in this project. The final data set has 4,424 cases and 39 variables. The purpose of this data is to predict whether or not a student will drop out of school. DropoutRisk is a binary classification variable with 2 classes (Dropout, Not_Dropout) which is the target variable used for modelling.

The original target variable consisted of three classes: Dropout, Enrolled, and Graduate. As Support Vector Machine is a binary classification algorithm, the target variable was converted into binary as the project is based on predicting whether the student is at risk to drop out or not. All students that were originally classified as "Dropout" were re-classed as Dropout, and students originally classified as "Graduate" and "Enrolled" were re-classed as Not_Dropout. This allowed us to directly concentrate on the prediction of dropout risk.

The information is made up of several variables pertaining to the education sector, like academic, demographic, financial, and enrolment variables. Relevant factors are admission grade, age at enrolment, being a scholarship holder, and keeping tuition fees current, approved course and curricular units during the first and second semester. These variables are helpful because they contain information about students' academic background and academic progress and can help to detect patterns associated with dropouts.

1.2 Data Dictionary

Variable	Description
DropoutRisk	Final Outcome variable to predict data. It can take one of two possible values, i.e., Dropout and Not_Dropout.
Target	Initial outcome variable with categories of Dropout, Enrolled, and Graduate. Excluded after creation of the DropoutRisk variable to avoid information leakage.
Partition	Variable used to create three mutually exclusive partitions of the data, namely, Training Set, Validation Set, and Testing Set.
Admission Grade	Admission grades of the student prior to joining the program.
Age at Enrollment	Age of the student when he/she joined the program.
Application Mode	Application method chosen by the student for applying to the program.

Application Order	Preference order for the chosen program within the application process.
Course	Academic program pursued by the student.
Scholarship	Scholarship holding status of the student.
Tuition fees up to date	Status indicating whether the tuition fees of the student were up-to-date or not.
Curricular Units 1st Semester Approved	Curricular units approved by the student in the 1st semester.
Curricular Units 2nd Semester Approved	Number of curricular units passed in the second semester.
Grade for Curricular Units 1st Semester	Academic grade attained in the first semester.
Grade for Curricular Units 2nd Semester	Academic grade attained in the second semester.
Student's Previous Qualification	The student's prior qualification attained before admission into the course.
Grade for Previous Qualification	Performance grades attained in the student's previous qualifications.
Gender	Student's Gender.
Mother's Occupation	The occupational class of the student's mother.
Unemployment Rate	Economic performance measures depict unemployment levels.

2.0 Business Case

2.1 Problem Statement / Business Goal

Student attrition is a critical issue for educational institutions, as it can have a negative impact on both the students and the institutions themselves. The project is intended to build models for the prediction of the probability of student dropouts, based on demographic, academic, and financial attributes. The objective analysis is to classify students into Dropout and Not_Dropout. The problem is a binary attribute; thus, it is a classification problem. The results of the analysis will consist of analyzing multiple classification models developed within SAS Viya Model Studio.

2.2 Aim and Specific Objectives

This assignment focuses on creating models that can predict student dropouts using the **STUDENT_DROPOUT_FINAL** dataset within SAS Viya.

Objectives include:

1. Exploring the student dropout dataset with an understanding of its structure, target variable distribution, and relevant input variables.
2. Preparing the dataset through creation of a binary target variable, rejecting the existing target variable, and performing data partitioning.
3. Performing variable selection on the dataset to determine the most valuable input variables for modeling.
4. Constructing six different predictive models using SAS Viya Model Studio. The models being built are:

    a. Neural Network
    b. Support Vector Machine
    c. Decision Tree
    d. Forest
    e. Logistic Regression
    f. Gradient Boosting

5. Optimizing and validating the models utilizing fit statistics, misclassification rate, accuracy, ROC, and model comparison findings.

6. Comparing all models and determining the **champion model** using the SAS Viya criteria for model comparison.
7. Interpret the models made and explain how these models can support student dropout risk predictions.

2.3 Scope of the Project

To predict students, drop-out risk the project uses the STUDENT_DROPOUT_FINAL dataset (4424 observations, 39 variables) within SAS Viya for Learners. DropoutRisk is the target variable where students are categorized as Dropout or Not_Dropout. The project consists of the development and a comparison of 6 predictive models that have not been deployed. Mohib works on Neural Network and SVM models, Zeba works on Decision Tree and Forest models, and Danish works on Logistic Regression and Gradient Boosting models with the final comparison being done together.

2.4 Methodology Selection

It will be carried out with the help of Cross Industry Standard Process for Data Mining (CRISP-DM). This project employs the CRISP-DM process as it is a systematic procedure for data mining problem solving that begins with the understanding of the problem in the business and ends with the evaluation of the model obtained and the interpretation of the model. The student will also need to describe the business objective, create data, build models, test models, and analyze the results as part of the assignment (performing the CRISP-DM process).

The initial step is Business Understanding. The biggest business issue faced with this project is the drop-out of students. The aim is to forecast students' dropouts and intervention if students are at risk of dropping out.

This second phase is referred to as Data Understanding. SAS Viya (Advanced Analytics) was used to explore the data set to understand the number of observations, number of variables, roles, distribution of the target variable, and levels of measurement. This enabled the group to understand the structure of the data set before modeling.

The third step is Data Preparation. The following steps were taken to prepare the final data set for this phase: Rejection of the original target variable to ensure that there is no leakage of information, configuration of the partition variable, and use of variable selection.

Modelling is the fourth stage. The models developed were Neural Network, SVM, Decision Tree, Forest, Logistic Regression, and Gradient Boosting. Each group member designed two models and tweaked the parameters of their model to get them to perform better.

Evaluation is the 5th step. The accuracy, misclassification rate, ROC, KS statistics, sensitivity, and specificity are some of the model evaluation measures. The accuracy, misclassification rate, ROC, KS statistics, sensitivity and specificity, and model comparison are some of the model evaluation measures. The aim of this stage was to determine which one of the models outperforms the test partition.

The last one is Deployment. This assignment is not initiated. Finally, the chosen model could be applied as a decision support system in educational institutions to determine the students who need extra academic and/or financial assistance.

3.0 Implementation - Analytical Solution

3.1 Exploratory Data Analysis

While developing predictive models, exploratory data analysis was conducted to understand the STUDENT_DROPOUT_FINAL dataset's structure, target variable, and substantial input factors. Before creating machine learning models, this stage is significant as it helps in determining the quantity of data, variables, target distribution, variable roles, and possible modeling issues.

There were 39 variables and 4,424 observations in the final Dataset. DropoutRisk , the project's target variable, classifies students into two groups, such as Dropout and Not_Dropout. Academic, demographic, economical and enrollment-based characteristics which could impact the likelihood of student dropout were also involved in the dataset.

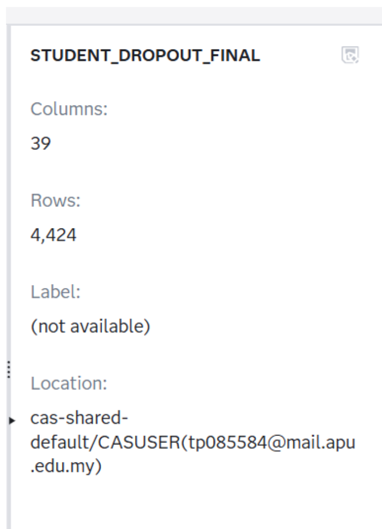


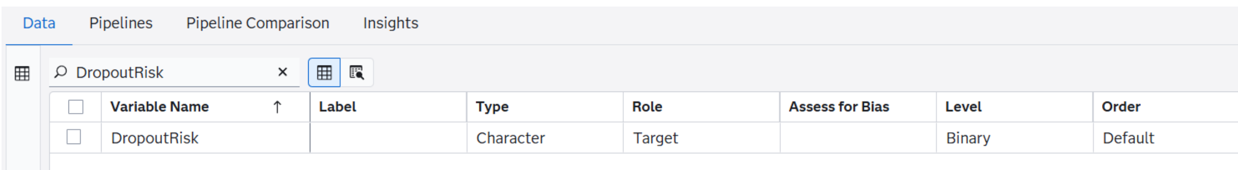
Figure 1: Imported STUDENT_DROPOUT_FINAL Dataset in SAS Viya

The imported STUDENT_DROPOUT_FINAL dataset in SAS viya is shown in figure 1. The final generated dataset was successfully loaded into the project, as demonstrated by the dataset's 4,424 rows and 39 columns.

Variable Name ↑	Label	Type	Role	Assess for Bias	Level	Order	Comment
PartInd		Numeric	Partition		Nominal	Default	
Admission grade		Numeric	Input		Interval	Default	
Age at enrollment		Numeric	Input		Interval	Default	
Application mode		Numeric	Input		Nominal	Default	
Application order		Numeric	Input		Nominal	Default	
Course		Numeric	Input		Nominal	Default	
Curricular units 1st sem (approv		Numeric	Input		Interval	Default	
Curricular units 1st sem (credit		Numeric	Input		Interval	Default	
Curricular units 1st sem (enroll		Numeric	Input		Interval	Default	

Figure 2: Variable Roles and Measurement Levels in the Dataset

The dataset's variable roles and measurement levels are demonstrated in Figure 2. PartInd has been classified as the partition variable, and most of the variables were classified as input variables. Based on the type of data, both nominal and interval variables are included in the measurement levels.

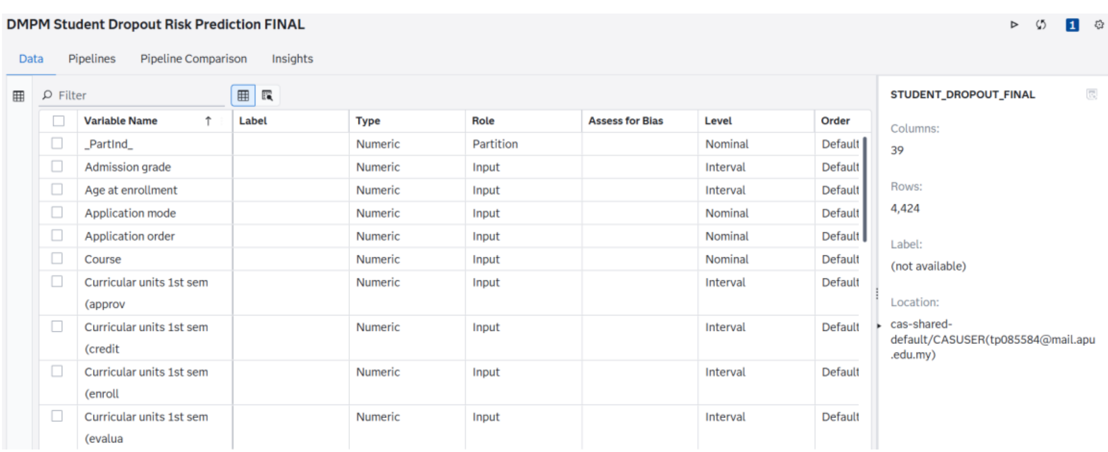


Variable Name	Label	Type	Role	Assess for Bias	Level	Order
DropoutRisk		Character	Target		Binary	Default

Figure 3: DropoutRisk Set as the Target Variable

DropoutRisk was properly classified as the goal variable as shown in Figure 3. This is crucial as this target variable is used via every predictive models to figure out a student's likelihood of dropping out.

3.1.1 Dataset Structure



Variable Name	Label	Type	Role	Assess for Bias	Level	Order
PartInd		Numeric	Partition		Nominal	Default
Admission grade		Numeric	Input		Interval	Default
Age at enrollment		Numeric	Input		Interval	Default
Application mode		Numeric	Input		Nominal	Default
Application order		Numeric	Input		Nominal	Default
Course		Numeric	Input		Nominal	Default
Curricular units 1st sem (approv		Numeric	Input		Interval	Default
Curricular units 1st sem (credit		Numeric	Input		Interval	Default
Curricular units 1st sem (enroll		Numeric	Input		Interval	Default
Curricular units 1st sem (evalua		Numeric	Input		Interval	Default

Figure 4: Dataset Structure Showing Rows and Columns

The STUDENT_DROPOUT_FINAL datasets structure in SAS Viya Model Studio is shown in Figure 4 . Academic, background, application, and financial variables are within 4,424 rows and 39 columns which make up the dataset. Since they help in the model's knowledge of possible impacts on a student's decision to proceed or discontinue their studies, these variables are useful for forecasting dropout risk. Both nominal and interval variables, which include grades, curriculum units, application mode, and course, are involved in the dataset.

3.1.2 Target Variable Distribution

The FREQ Procedure

DropoutRisk	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Dropout	1421	32.12	1421	32.12
Not_Dropout	3003	67.88	4424	100.00

Figure 5: DropoutRisk Target Distribution

The distribution of Dropoutrisk the final binary target variable is shown in Figure 5. The dataset is relatively unbalanced as demonstrated by the 1,421 Dropout students (32.12%) and 3,003 Not_Dropout students (67.88%).

Target	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Dropout	1421	32.12	1421	32.12
Enrolled	794	17.95	2215	50.07
Graduate	2209	49.93	4424	100.00

Figure 6: Original Target Variable Distribution

The original Target variable with three classes- Dropout (1,412), Enrolled (794), and Graduate (2,209) is shown in Figure 6. The objective turned into a binary variable where Dropout remained as Dropout and Enrolled as well as Graduated were classified as Not_Dropout to be able to focus on dropout risk prediction.

3.1.3 Important Input Variables

Variable Selection

Name	Variable Label	Variable Level	Role	Reason
AGE AT ENROLLMENT		INTERVAL	INPUT	
APPLICATION MODE		NOMINAL	INPUT	
COURSE		NOMINAL	INPUT	
CURRICULAR UNITS 1ST SEM (APPROV		INTERVAL	INPUT	
CURRICULAR UNITS 1ST SEM (CREDIT		INTERVAL	INPUT	
CURRICULAR UNITS 1ST SEM (WITHOU		NOMINAL	INPUT	
CURRICULAR UNITS 2ND SEM (APPROV		INTERVAL	INPUT	
CURRICULAR UNITS 2ND SEM (CREDIT		NOMINAL	INPUT	
CURRICULAR UNITS 2ND SEM (ENROLL		INTERVAL	INPUT	

Figure 7: Important Input Variables Identified by Variable Selection

The important variables used to predict DropoutRisk are shown in Figure 7. Academic achievement, enrollment information and student background, which includes age at enrollment, application manner, course, and first and second semester curriculum units are the most frequently used variables. These factors help the models in finding students who can be more likely to drop out.

3.1.4 Summary of EDA Findings

The exploratory data analysis figured that the final dataset is appropriate for predictive modeling. The STUDENT_DROPOUT_FINAL dataset has 4,424 observations and 39 variables, which is enough data for training, validation, and testing. The target variable DropoutRisk was properly set up as a binary variable with two classes, such as Dropout and Not_Dropout.

The target distribution found that 32.12% of students were classified as Dropout, while 67.88% were classified as Not_Dropout. This indicates a moderate class imbalance, but the dataset stays feasible for classification models. The actual Target variable was also examined to make sure the actual distribution of Dropout, Enrolled and Graduate students.

The EDA also found that various significant input variables relate to academic success. Certainly, curricular units accepted, credited, enrolled, and evaluated throughout the first and second semesters. These characteristics are essential as students' academic success has a strong association with dropout risk. Therefore, the EDA findings support the use of predictive modelling to identify students who are at risk of dropping out.

3.2 Data Cleaning and Pre-processing

The dataset was cleaned and pre-processed to prepare it for modeling in SAS Viya Model Studio. This stage included the dataset, creating the final binary target variable, denying the initial target variable to avoid information leaks, producing a data division variable, and using variable selection.

Pre-processing is significant as predictive models need a clean, well-organized dataset. If the target variable input roles, or partition parameters are wrong, the model might produce incorrect outcomes. As an outcome, the dataset was carefully examined before developing the six predictive models.

3.2.1 Dataset Import

The CONTENTS Procedure

Data Set Name	CASUSER.STUDENT_DROPOUT_FINAL	**Observations**	4424
Member Type	DATA	**Variables**	39
Engine	CAS	**Indexes**	0
Created	05/30/2026 13:45:21	**Observation Length**	320
Last Modified	05/30/2026 13:45:21	**Deleted Observations**	0
Protection		**Compressed**	NO
Data Set Type		**Sorted**	NO
Label			
Data Representation	SOLARIS_X86_64, LINUX_X86_64, ALPHA_TRU64, LINUX_IA64, LINUX_POWER_64		
Encoding	utf-8 Unicode (UTF-8)		

Figure 8: STUDENT_DROPOUT_FINAL Created Successfully in SAS Studio

STUDENT_DROPOUT_FINAL was successfully built in SAS studio by using 4,424 data and 39 variables. The dataset was saved in the CASUSER library and used as the shared dataset over all models.

3.2.2 Target Variable Preparation

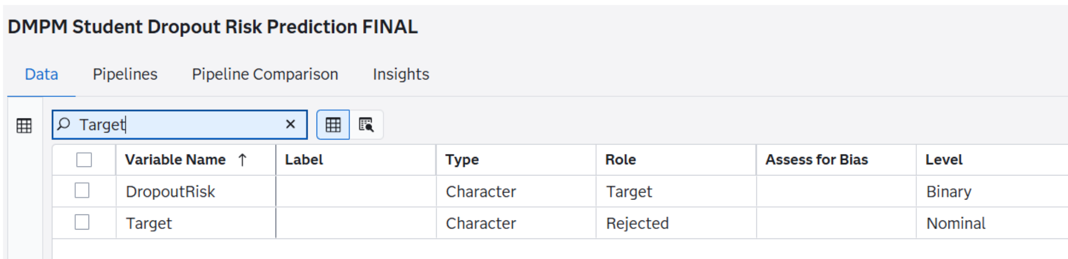
```
length DropoutRisk $12;

if Target = "Dropout" then DropoutRisk = "Dropout";
else DropoutRisk = "Not_Dropout";
```

Figure 9: Binary Target Variable DropoutRisk Preparation

This shows the SAS code used to explain DropoutRisk as a binary target variable. Dropout, remained as Dropout, while Enrolled and Grduate were classified as Not_Dropout, which results in dataset appropriate for binary classification.

3.2.3 Information Leakage Check



DMPM Student Dropout Risk Prediction FINAL

Data Pipelines Pipeline Comparison Insights

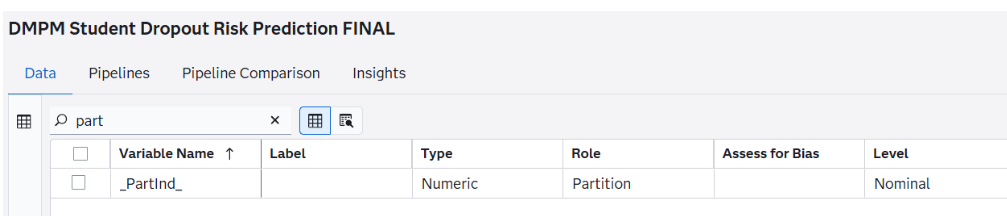
Target

Variable Name ↑	Label	Type	Role	Assess for Bias	Level
DropoutRisk		Character	Target		Binary
Target		Character	Rejected		Nominal

Figure 10: Original Target Variable Rejected to Prevent Information Leakage

This shows how the actual Target variable was denied avoiding information leaking. DropoutRisk was created from Target, thus employing Target as an input could make the model results unrealistic.

3.2.4 Data Partition



DMPM Student Dropout Risk Prediction FINAL

Data Pipelines Pipeline Comparison Insights

part

Variable Name ↑	Label	Type	Role	Assess for Bias	Level
PartInd		Numeric	Partition		Nominal

Figure 11: PartInd Set as the Partition Variable

This illustrates that PartInd was utilized as a partition variable to split the dataset into training, validation, and testing groups.

PartInd	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	1389	31.40	1389	31.40
1	2609	58.97	3998	90.37
2	426	9.63	4424	100.00

Figure 12: Training, Validation and Testing Partition Distribution

This shows the partition distribution as 2,609 training records (58.97%), 1,389 validation records (31.40%), and 426 testing records (9.63%). These divisions made sure that the models were equitably trained, tuned, and tested.

3.2.5 Variable Selection

Variable Selection was used employed in the pre-processing pipeline to find interesting input variables before building the prediction models. This step assists to reduce irrelevant variables, enabling the model to emphasize the predictors which are most relevant to the target variable DropoutRisk.

The Variable Selection node was added after the Data node was in the pipeline. The chosen variables were then given to all six models, making sure that each model used identical pre-processed input data. This makes the model contrast fair as each model is trained with the same set of variables.

Various academic and enrollment-based parameters were selected as input variables, which include curricular units which were authorized, credited, and enrolled in the first and second semesters. These variables are helpful as they imply student's academic achievement, which is strongly associated with dropout risk.

3.2.6 Final Pre-processing Pipeline

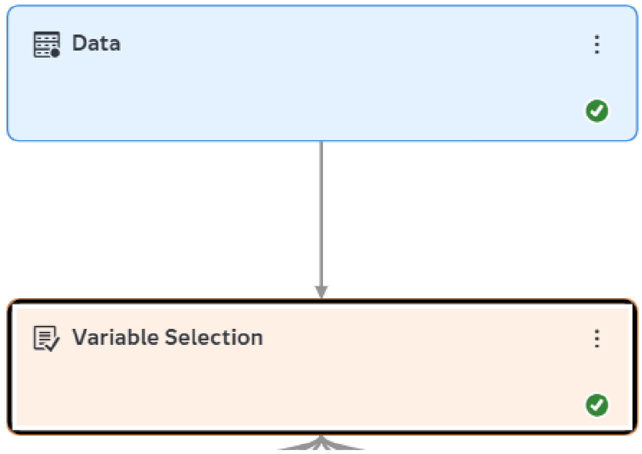


Figure 13: Final Shared Pre-processing Pipeline

This shows the final usual pre-processing pipeline in SAS Viya Model Studio. The Data and Variable Selection nodes were successfully accomplished, and the selected variables were employed by all models to make sure of a fair comparison.

3.3 Model Construction and Optimization

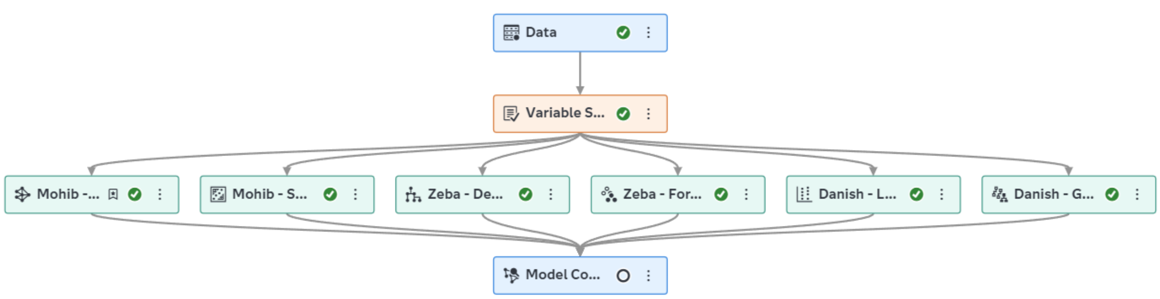


Figure 14: Final SAS Viya Pipeline with All Six Predictive Models

3.3.1 Mohib - Neural Network Model

A Neural Network is a supervised machine learning model built around the architecture of the human brain. It discovers trends from data through interconnected nodes and is highly suitable for tackling classification issues with complex variable interactions. In this project, a Neural Network model was employed to forecast the binary target variable DropoutRisk, which labels students as Dropout or Not Dropout.

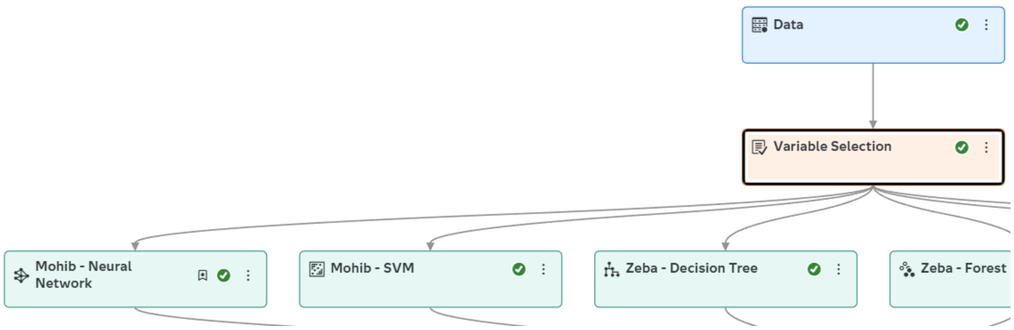


Figure 15: Mohib - Neural Network Node

This figure shows the Mohib – Neural Network node attached to the shared pre-processing pipeline. This indicates that the Neural Network model was developed using the same prepared datasets and inputs as the other models in the project. Using the same pipeline helps ensure that the model evaluations are fair and accurate.

The SAS System

The NNET Procedure

Model Information	
Model	Neural Net
Number of Observations Used	2609
Number of Observations Read	2609
Target/Response Variable	DropoutRisk
Number of Nodes	142
Number of Input Nodes	90
Number of Output Nodes	2
Number of Hidden Nodes	50
Number of Hidden Layers	1
Number of Weight Parameters	4550
Number of Bias Parameters	52
Architecture	MLP
Seed for Initial Weight	12345
Optimization Technique	LBFGS
Number of Neural Nets	1
Objective Value	1.6490747159
Misclassification Rate for Validation	0.1141

Figure 16: Baseline Neural Network Model Information

This figure displays the baseline Neural Network model information. The baseline model features a Multi-Layer Perceptron design with one hidden layer and fifty hidden nodes. The model's goal variable was DropoutRisk, and it was trained with input parameters from the same initial processing. The baseline model had a validation misclassification rate of 0.1159, indicating that it correctly detected roughly 88.41% of the validation observations.

Multiple tuning effects were made to enhance the performance of the Neural Network. The number of hidden nodes rose from 50 to 75 or decreased from 50 to 25, and an extra hidden layer was attempted. Activation functions like Logistic and ReLU were also explored. These tuning procedures were carried out to figure out whether changing the network structure will boost model performance.

Fit Statistics

Target ...	Data Role	Partitio...	Misclas...
DropoutRisk	TEST	2	0.2958
DropoutRisk	TRAIN	1	0.3281
DropoutRisk	VALIDATE	0	0.3161

Figure 17: Neural Network Tuning Result with 25 Hidden Nodes

Fit Statistics

Target ...	Data Role	Partitio...	Formatt...	Misclas...
DropoutRisk	TEST	2	2	0.1056
DropoutRisk	TRAIN	1	1	0.1284
DropoutRisk	VALIDATE	0	0	0.1159

Figure 18: Neural Network Tuning Result with 75 Hidden Nodes

Fit Statistics

Target ...	Data Role	Partitio...	Formatt...	Misclassification Rate (Event)
DropoutRisk	TEST	2	2	0.2958
DropoutRisk	TRAIN	1	1	0.3281
DropoutRisk	VALIDATE	0	0	0.3161

Figure 19: Neural Network Tuning Result with Two Hidden Layers

The tuning result figures indicate that altering the Neural Network settings did not optimize the model. When 25 hidden nodes were evaluated, the validation misclassification rate climbed to 0.3161, more than the baseline model. When 75 hidden nodes were examined, the validation misclassification rate remained at 0.1159, which had no impact on the baseline outcome. Using two hidden layers gave a lower result, at a validation misclassification result of 0.3161. The Logistic and ReLU activation function evaluations did not enhance the model performance.

Neural Network Tuning Comparison:

Neural Network Version	Setting Tested	Validation Misclassification	Decision
Baseline	1 hidden layer, 50 hidden nodes	0.1159	Selected
Tuned 1	1 hidden layer, 25 hidden nodes	0.3161	Not selected
Tuned 2	1 hidden layer, 75 hidden nodes	0.1159	Not selected
Tuned 3	2 hidden layers, 50 hidden nodes	0.3161	Not selected
Tuned 4	Logistic/ReLU activation tested	0.3161	Not selected

The Neural Network tuning comparison table combines all tuning efforts. Based on the analysis, the baseline design with one hidden layer and 50 hidden nodes was determined as the final Neural Network model. This is because it generated the most precise and accurate validation outcomes in comparison with the other tuned variants.

The results disclose that the Neural Network was originally fully optimized for the baseline design. The model did not need any further complexity because raising the number of hidden layers or altering activation functions decreased performance rather than enhanced. As a result, the chosen Neural Network model is seen adequate and stable for this dataset.

3.3.2 Mohib - Support Vector Machine Model

Support Vector Machine (SVM) is a supervised machine learning method used for grouping tasks. SVM tries to determine the separation border between classes. In this instance, SVM was acceptable because the target variable DropoutRisk was converted into a binary target consisting of Dropout and Not_Dropout.

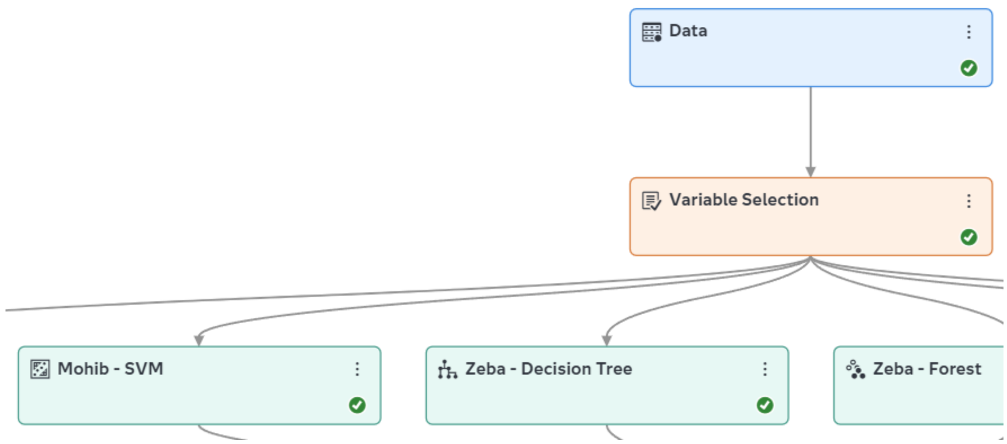


Figure 20: Baseline Mohib - SVM Node

This figure shows the connection between the shared pre-processing pipeline and the baseline Mohib – SVM node. This suggests that the SVM model used the same processed dataset and variables as the rest of the models. This is essential as all models must use similar cleaned data for an accurate comparison.

The SAS System

The SVMACHINE Procedure

Model Information	
Task Type	C_CLAS
Optimization Technique	Interior Point
Scale	YES
Kernel Function	Linear
Penalty Method	C
Penalty Parameter	1
Maximum Iterations	25
Tolerance	1e-06

Figure 21: Baseline SVM Model Information

This figure depicts the baseline SVM model data. The baseline SVM has a linear kernel, with a penalty parameter set to one. DropoutRisk was utilized as the model target variable, and the input variables were selected from the shared pre-processing pipeline for training purposes. The baseline SVM had a validation misclassification rate of 0.1166, meaning it accurately identified about 88.34% of the validation data.

Fit Statistics

Target ...	Data Role	Partitio...	Formatt...	Misclassification Rate (Event)
DropoutRisk	TEST	2	2	0.1009
DropoutRisk	TRAIN	1	1	0.1276
DropoutRisk	VALIDATE	0	0	0.1152

Figure 22: SVM Tuning Result with Penalty Parameter 0.5

Fit Statistics

Target ...	Data Role	Partitio...	Formatt...	Misclassification Rate
DropoutRisk	TEST	2	2	0.1080
DropoutRisk	TRAIN	1	1	0.1219
DropoutRisk	VALIDATE	0	0	0.1166

Figure 23: SVM Tuning Result with Penalty Parameter 2

Several penalty parameter values were examined to enhance the SVM model performance. The penalty parameter defines how harshly the SVM Model responds to classification errors. A reduced penalty permits for a wider margin, whereas an increased penalty seeks to restrict training errors more firmly. Three variants were examined: the baseline penalty value of one, a tuned penalty value of 0.5, and a tuned penalty value of two.

The tuning result with a penalty value of 0.5 gave the optimal SVM performance. It lowered the validation misclassification rate from 0.1166 to 0.1152. This demonstrates a slight but substantial improvement above the baseline SVM model. The tuning result with a penalty parameter of 2 failed to improve the model because the validation misclassification rate remained stable at 0.1166, thus being identical to the baseline result.

SVM Tuning Comparison:

SVM Version	Kernel	Penalty Parameter	Validation Misclassification	Decision
Baseline	Linear	1	0.1166	Not selected
Tuned 1	Linear	0.5	0.1152	**Selected**
Tuned 2	Linear	2	0.1166	Not selected

The SVM tuning comparison table offers an overview of all assessed penalty levels. According to the comparison results, the final SVM model implemented a linear kernel with a penalty parameter of 0.5. This result was decided because it generated the lowest validation misclassification rate out of the assessed SVM variants.

In general, the SVM model was efficient in forecasting student dropout risk. Despite the small enhancement, the penalty parameter 0.5 gave the best results. Hence, this setting was chosen as the final tuned SVM model.

3.3.3 Zeba - Decision Tree Model

Decision trees form examples of supervised machine learning methods capable of carrying out classifications and regressions. They display decisions and outcomes in a tree structure whereby each node of the tree splits data based on some specific criteria. The decision trees have simple concepts, require minimal data pre-processing, and aid in identifying important predictors.

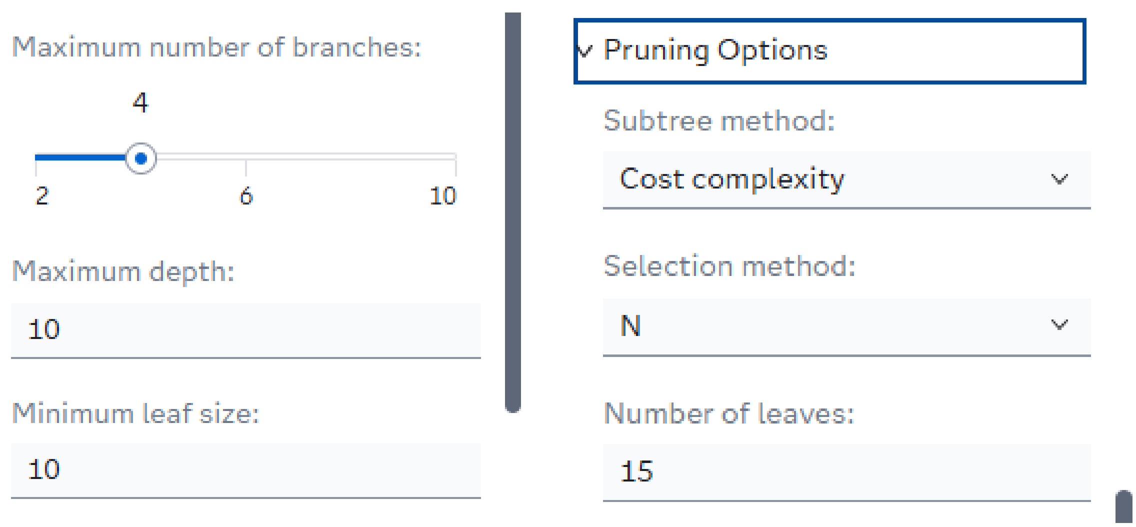


Figure 24: Tuned Decision Tree settings

Before building this model, few changes were made to the splitting and pruning options. The maximum branch had been changed to 4. The number of leaves increased to 15 and the selection method changed to N.

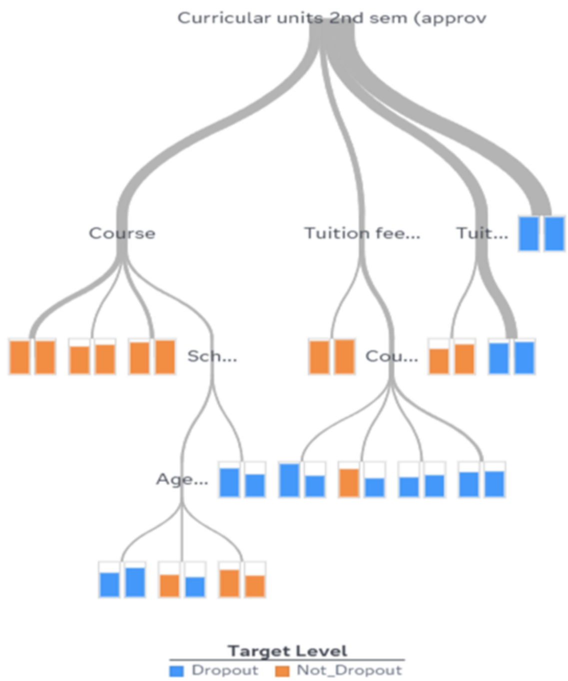


Figure 25: Decision Tree

3.3.3.1 Branch 2, N:12

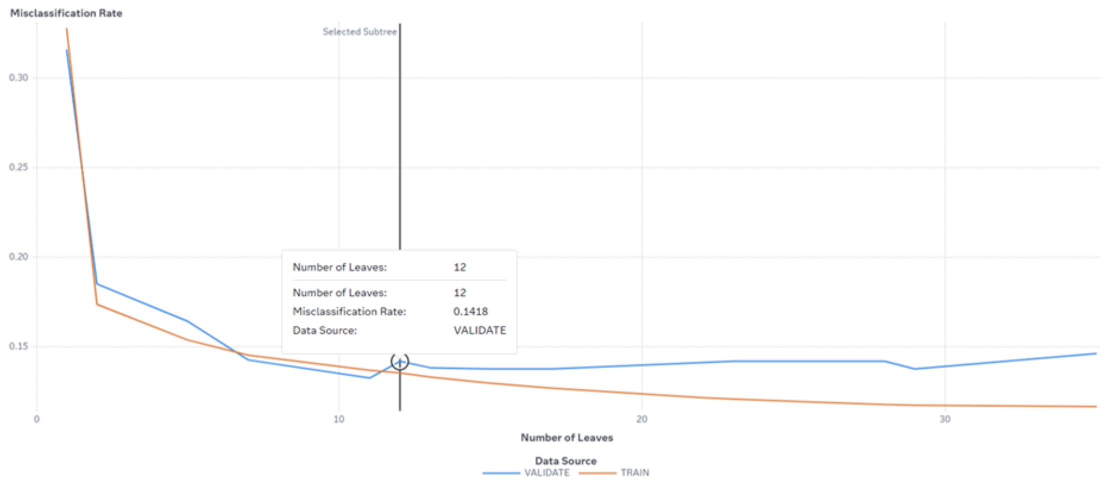


Figure 26: Validation Misclassification Rate (Baseline)

Misclassification Rate (Validate) - 0.1418

3.3.3.2 Branch 4, N:15

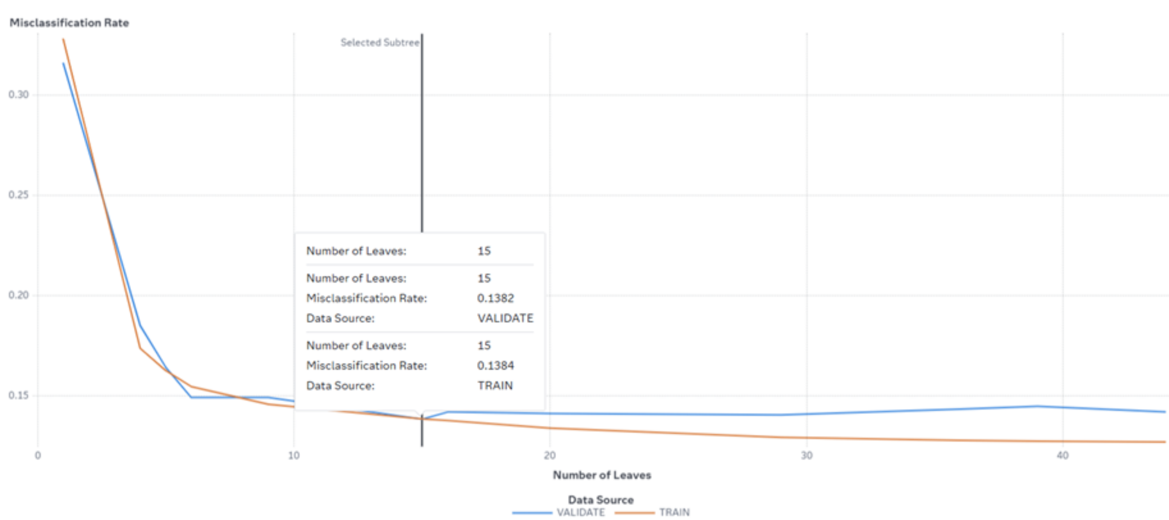


Figure 27: Validation Misclassification Rate (Tuned)

Misclassification Rate (Validate) - 0.1382

3.3.3.3 Branch 6, N:17

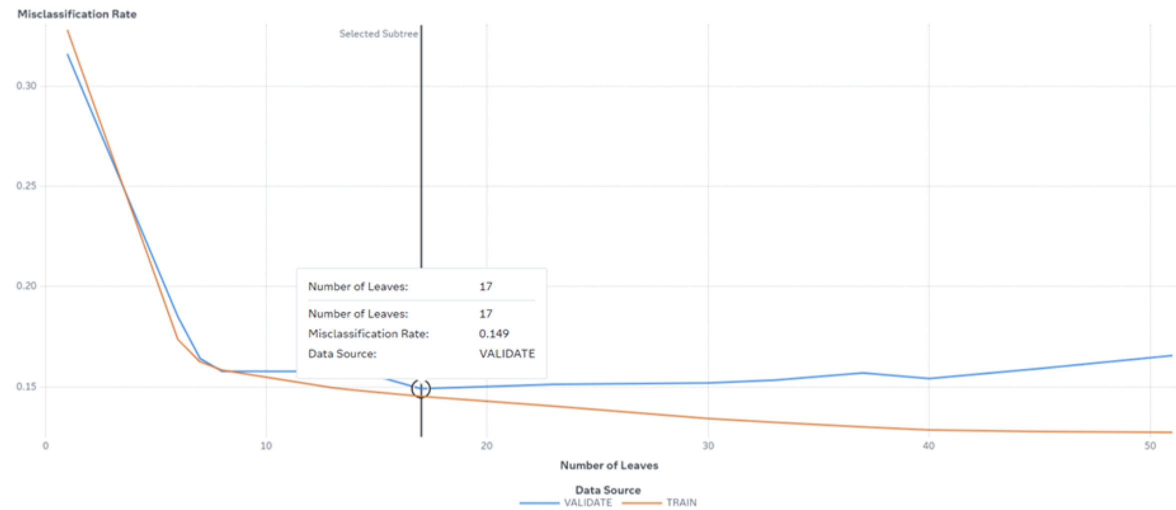


Figure 28: Validation Misclassification Rate (Baseline)

Misclassification Rate (Validate) - 0.149

Branch 4 has the lowest misclassification rate when compared to branch 2 and branch 6. Therefore, the max branch was changed to 4.

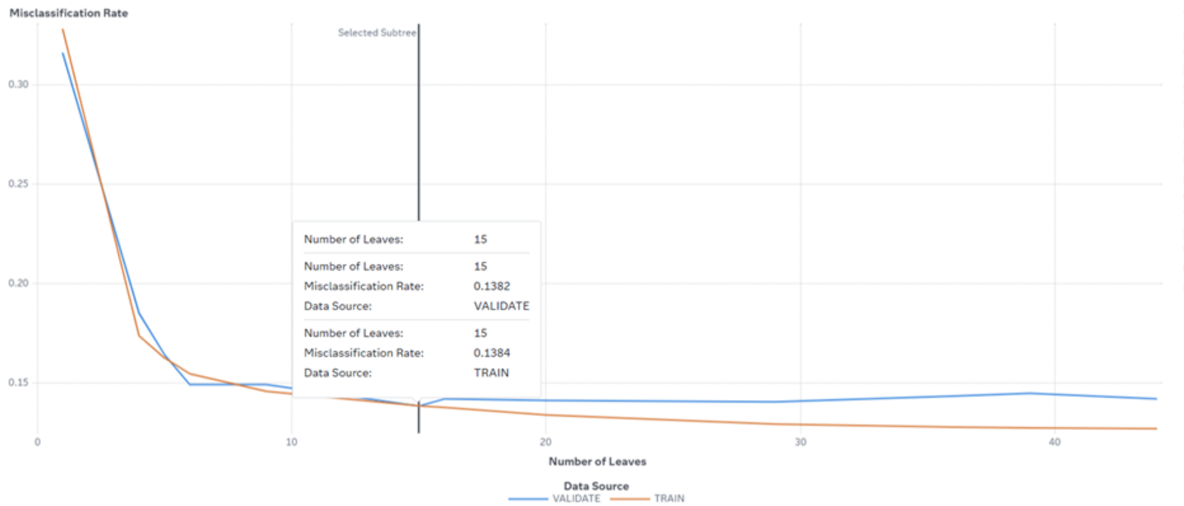


Figure 29: Misclassification Rate Graph

The selection of the decision tree with 15 leaves is appropriate because it produced a small validation misclassification error of 0.1382 which is almost equal to the training misclassification error of 0.1384. The similarity between validation and training misclassification error indicates that the model does not suffer from over-fitting. Also, increasing the number of leaves beyond 15 will not lead to any remarkable decrease in the validation error but will only increase complexity.

3.3.4 Zeba - Forest Model

Forest model is one of many machine learning algorithms and data mining techniques whose principle of operation lies in the use of the ensemble of decision trees. Combining decisions of several trees allows achieving higher efficiency through more accurate prediction results, overcoming problems such as overfitting, and processing a large amount of data.

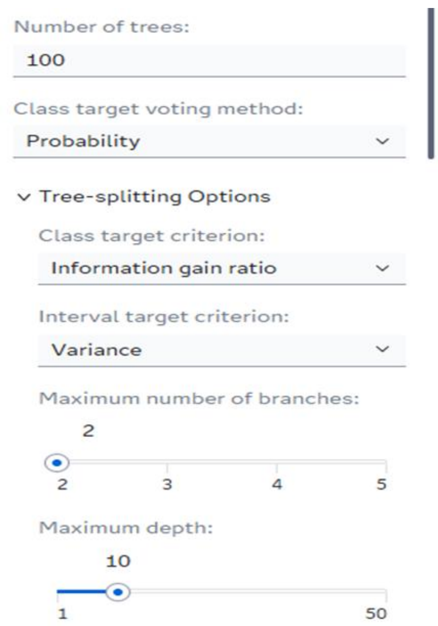


Figure 30: Tuned Forest Model Settings

Forest Model configuration included 100 trees, maximum tree depth equal to 10, and the maximum number of branches being equal to two to find the optimal balance between predictive power and efficiency while limiting the risk of overfitting.

Figure 31: Misclassification Rate Graph showing four data sources

All four lines present a sharp decrease in misclassification rate at the initial phase of building a Forest Model and then continue the trend of stable values, implying that convergence occurs in this range and further increase in the number of trees does not lead to improvements in performance. At the same time, the TEST line demonstrates the lowest misclassification rate of about 0.10, followed by TRAIN rate reaching 0.12 and remaining at these levels. For VALDIDATE and OUT OF BAG datasets, the misclassification rate ranges from 0.12 to 0.13 and from 0.13 to 0.14, respectively. Therefore, the Forest model with 100 trees is considered a stable and well-fitted model for predicting student dropouts.

3.3.5 Danish - Logistic Regression Model

Logistic Regression is a supervised machine learning algorithm commonly used for classification problems. It predicts the probability that an observation belongs to a particular class by modelling the relationship between the target variable and the predictor variables. In this study, Logistic Regression was used to predict student dropout risk based on academic, demographic, and socio-economic factors. The target variable, DropoutRisk, consists of two classes: Dropout and Not_Dropout.

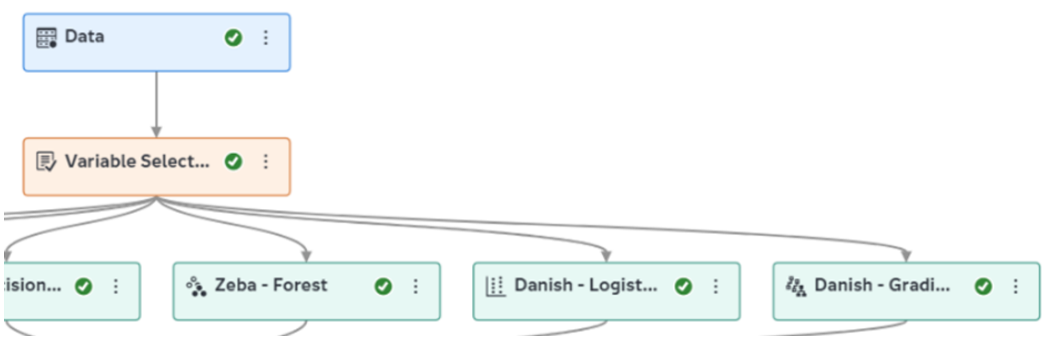


Figure 32: Basic Danish - Linear Regression Node

This figure shows the Logistic Regression node attached to the shared Variable Selection node within the SAS Viya pipeline. The model was developed using the same pre-processed dataset, partition settings, and selected input variables as the other predictive models in the project. Using the same preprocessing pipeline ensures consistency and fairness when comparing model performance, as all models are trained and validated using identical data preparation procedures. This allows the evaluation results to accurately reflect the strengths and weaknesses of the Logistic Regression model rather than differences in data processing.

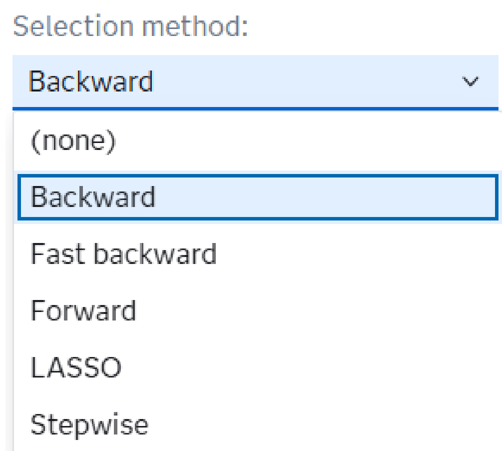
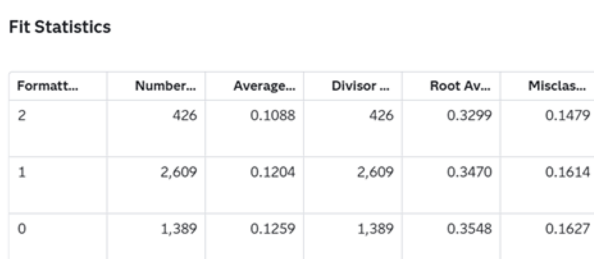


Figure 33: Tuned Logistic Regression Settings

Before building the final model, three variable selection methods were evaluated: Stepwise Selection, Forward Selection, and Backward Elimination. These methods were compared using validation misclassification rate to determine the best-performing model.



Fit Statistics

Formatt...	Number...	Average...	Divisor ...	Root Av...	Misclas...
2	426	0.1088	426	0.3299	0.1479
1	2,609	0.1204	2,609	0.3470	0.1614
0	1,389	0.1259	1,389	0.3548	0.1627

Figure 34: Validation Misclassification Rate (Stepwise)

Misclassification Rate (Train) = 0.1479

Misclassification Rate (Validate) = 0.1614

Misclassification Rate (Test) = 0.1627

Fit Statistics

Formatt...	Number...	Average...	Divisor ...	Root Av...	Misclas...	Multi-Cl...	KS (You...
2	426	0.1088	426	0.3299	0.1479	0.3554	0.6583
1	2,609	0.1204	2,609	0.3470	0.1614	0.3908	0.6425
0	1,389	0.1259	1,389	0.3548	0.1627	0.4293	0.6297

Figure 35: Validation Misclassification Rate (Forward)

Misclassification Rate (Train) = 0.1479

Misclassification Rate (Validate) = 0.1614

Misclassification Rate (Test) = 0.1627

Fit Statistics

Formatt...	Number...	Average...	Divisor ...	Root Av...	Misclas...	Multi-Cl...	KS (You...
2	426	0.0820	426	0.2863	0.1033	0.2778	0.7505
1	2,609	0.0993	2,609	0.3152	0.1307	0.3327	0.6970
0	1,389	0.0941	1,389	0.3068	0.1188	0.3188	0.7011

Figure 36: Validation Misclassification Rate (Backward)

Misclassification Rate (Train) = 0.1033

Misclassification Rate (Validate) = 0.1307

Misclassification Rate (Test) = 0.1188

The tuning results indicate that the Backward Elimination method produced the best Logistic Regression model. Both Stepwise and Forward Selection generated identical validation misclassification rates of 0.1614. In comparison, Backward Elimination achieved a lower validation misclassification rate of 0.1307, representing an improvement in predictive performance.

Logistic Regression Tuning Comparison:

Logistic Regression Version	**Selection Method**	**Validation Misclassification**	**Decision**
Tuned 1	Stepwise	0.1614	Not Selected
Tuned 2	Forward	0.1614	Not Selected
Tuned 3	Backward Elimination	0.1307	Selected

The Logistic Regression tuning comparison table summarizes all tested selection methods. Based on the comparison results, the final Logistic Regression model used the Backward Elimination selection method because it produced the lowest validation misclassification rate among all evaluated models.

The selected model achieved a training misclassification rate of 0.1033 and a validation misclassification rate of 0.1307. The difference between these values is only 0.0274 (2.74%), which indicates that the model does not suffer from overfitting. Furthermore, the validation accuracy of the selected model was 86.93%, while the test accuracy reached 88.12%. These results demonstrate that the Logistic Regression model is stable, well-fitted, and effective in predicting student dropout risk.

3.3.6 Danish - Gradient Boosting Model

Gradient Boosting is an ensemble machine learning technique that builds multiple decision trees sequentially, where each new tree attempts to correct the errors made by the previous trees. The model combines the predictions of many weak learners to create a stronger predictive model. In this project, Gradient Boosting was used to predict the binary target variable DropoutRisk, which classifies students as either Dropout or Not_Dropout.

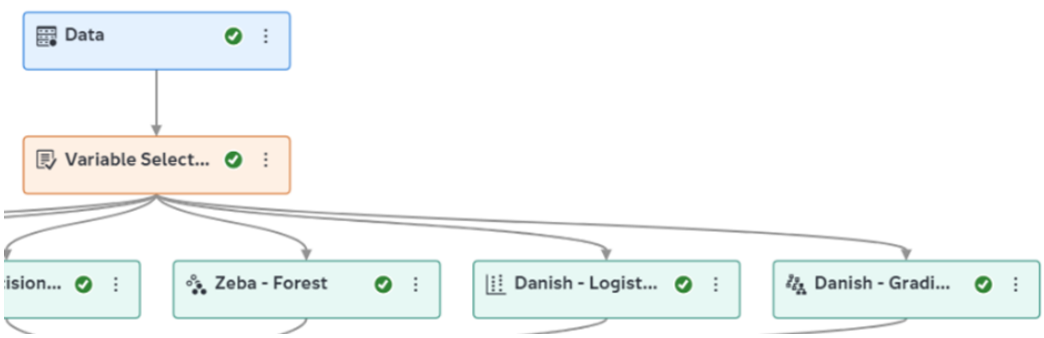


Figure 37: Baseline Danish – Gradient Boosting Node

This figure shows the Gradient Boosting node attached to the shared Variable Selection node within the SAS Viya pipeline. The model was developed using the same pre-processed dataset and selected variables as the other predictive models. This ensures consistency and allows fair comparison between all models in the project.

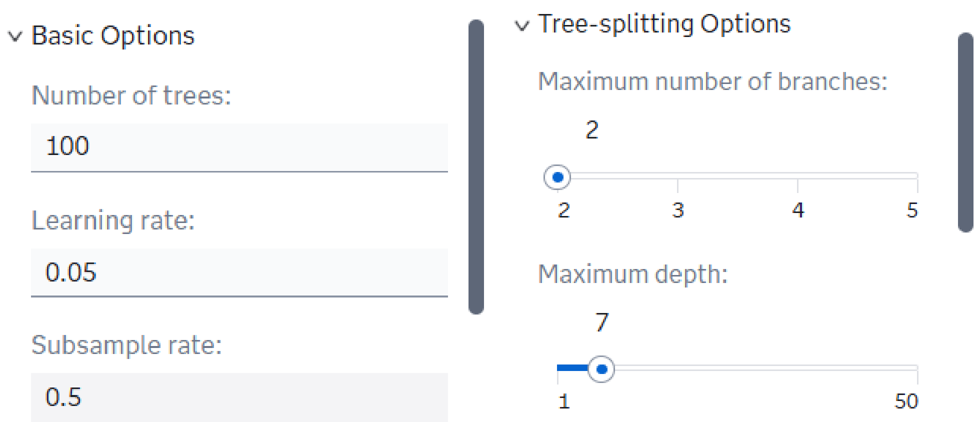


Figure 38: Tuned Gradient Boosting Settings

Several Gradient Boosting configurations were evaluated by adjusting the number of trees, learning rate, and maximum tree depth. These parameters were tuned to identify the model that produced the lowest validation misclassification rate while maintaining good generalization performance

Default Configuration

Misclassification Rate (Train) = 0.0920

Misclassification Rate (Validate) = 0.1253

Misclassification Rate (Test) = 0.1009

50 Trees, Learning Rate 0.1, Depth 3

Misclassification Rate (Train) = 0.1345

Misclassification Rate (Validate) = 0.1231

Misclassification Rate (Test) = 0.0962

100 Trees, Learning Rate 0.1, Depth 5

Misclassification Rate (Train) = 0.0751

Misclassification Rate (Validate) = 0.1260

Misclassification Rate (Test) = 0.1174

150 Trees, Learning Rate 0.1, Depth 5

Misclassification Rate (Train) = 0.0751

Misclassification Rate (Validate) = 0.1260

Misclassification Rate (Test) = 0.1174

100 Trees, Learning Rate 0.05, Depth 5

Misclassification Rate (Train) = 0.3281

Misclassification Rate (Validate) = 0.3161

Misclassification Rate (Test) = 0.2958

100 Trees, Learning Rate 0.05, Depth 7

Misclassification Rate (Train) = 0.3281

Misclassification Rate (Validate) = 0.3161

Misclassification Rate (Test) = 0.2958

Among all evaluated configurations, the model with 50 trees, learning rate 0.1, and maximum depth 3 achieved the lowest validation misclassification rate of 0.1231. Therefore, this configuration was selected as the final Gradient Boosting model.

The selected model achieved a validation accuracy of 87.69%. The difference between the training misclassification rate (0.1345) and validation misclassification rate (0.1231) was only 1.14%, indicating that the model generalizes well and does not suffer from overfitting. In contrast, the configurations using a learning rate of 0.05 produced substantially higher misclassification rates and lower accuracy, suggesting underfitting. Based on these results, the selected Gradient Boosting model is considered stable, well-fitted, and effective for predicting student dropout risk.

3.4 Model Validation

3.4.1 Neural Network Validation

The final Neural Network model was assessed by the fit statistics and misclassification rates from the validation, training, and testing parts. The testing misclassification rate was 0.1033, suggesting that the model accurately detected about 89.67% of testing data.

Score Information for Training	
Number of Observations Read	2609
Number of Observations Used	2609
Misclassification Rate	0.1269

Score Information for Validation	
Number of Observations Read	1389
Number of Observations Used	1385
Misclassification Rate	0.1141

Score Information for Testing	
Number of Observations Read	426
Number of Observations Used	426
Misclassification Rate	0.1033

Figure 39: Baseline Neural Network Score Information

The final Neural Network fit statistics reveals that the validation, training, and testing misclassification rates are 0.1159, 0.1269, and 0.1033, correspondingly. These same outcomes prove that the model predicts accurately and displays no indications of significant overfitting.

Fit Statistics

Target ...	Data Role	Partitio...	Formatt...	Number...	Average...	Divisor ...	Root Av...	Misclas...	Multi-Cl...	KS (You...	Area Un...	Gini Coe...	Gamma
DropoutRisk	TEST	2	2	426	0.0808	426	0.2843	0.1033	0.2805	0.7730	0.9358	0.8717	0.8762
DropoutRisk	TRAIN	1	1	2,609	0.0954	2,609	0.3088	0.1269	0.3188	0.7197	0.9239	0.8478	0.8526
DropoutRisk	VALIDATE	0	0	1,389	0.0952	1,389	0.3085	0.1159	0.3233	0.7053	0.9158	0.8316	0.8366

Figure 40: Final Neural Network Fit Statistics

Excellent classification accuracy was proven by the testing partitions Area Under the ROC Curve (AUC) of 0.9358. SAS Viya chose the Neural Network as the champion model given that it also achieved the maximum KS value in the final comparison of the models.

In general, the Neural Network exhibited outstanding predictive performance and was efficient in recognizing students who might be at risk of dropping out.

3.4.2 SVM Validation

A Linear Kernel having a penalty parameter of 0.5 was implemented in the final optimum SVM model. At a validation misclassification rate of 0.1152, it barely surpassed the baseline rate of 0.1166.

Fit Statistics			
Statistic	Training	Validation	Testing
Accuracy	0.8739	0.8834	0.8991
Error	0.1261	0.1166	0.1009
Sensitivity	0.9509	0.9484	0.9567
Specificity	0.7161	0.7426	0.7619

Figure 41: Baseline SVM Fit Statistics

With training, validation, and testing accuracy levels of 87.39%, 88.34%, 89.91%, correspondingly, the baseline SVM initially showed strong performance. Outstanding results on unknown data were illustrated by the testing misclassification rate of 0.1009.

Misclassification Matrix									
	Training Prediction			Validation Prediction			Testing Prediction		
Observed	Not_Dropout	Dropout	Total	Not_Dropout	Dropout	Total	Not_Dropout	Dropout	Total
Not_Dropout	1667	86	1753	901	49	950	287	13	300
Dropout	243	613	856	113	326	439	30	96	126
Total	1910	699	2609	1014	375	1389	317	109	426

Figure 42: Baseline SVM Misclassification Matrix

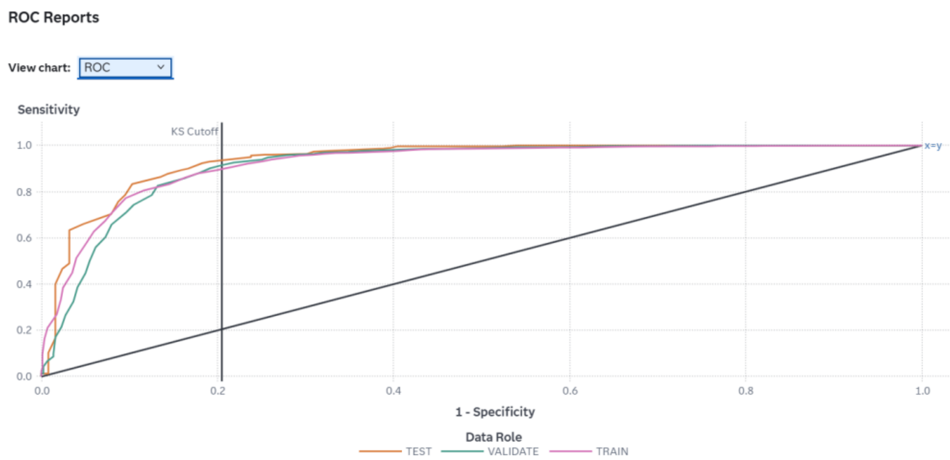


Figure 43: Baseline SVM ROC Curve

While many dropout students continued to be incorrectly detected, the baseline misclassification matrix demonstrated that a majority of Dropout and Not_Dropout cases were properly categorized. Substantial class separation was further shown by the ROC Curve, which had a testing AUC of roughly 0.9372.

Fit Statistics			
Statistic	**Training**	**Validation**	**Testing**
Accuracy	0.8724	0.8848	0.8991
Error	0.1276	0.1152	0.1009
Sensitivity	0.9498	0.9495	0.9567
Specificity	0.7138	0.7449	0.7619

Figure 44: Final Tuned SVM Fit Statistics

Misclassification Matrix									
	Training Prediction			Validation Prediction			Testing Prediction		
Observed	Not_Dropout	Dropout	Total	Not_Dropout	Dropout	Total	Not_Dropout	Dropout	Total
Not_Dropout	1665	88	1753	902	48	950	287	13	300
Dropout	245	611	856	112	327	439	30	96	126
Total	1910	699	2609	1014	375	1389	317	109	426

Figure 45: Final Tuned SVM Misclassification Matrix

Immediately after the tuning, the testing accuracy stood at 89.91% whereas the validation accuracy improved to 88.48%. The tuned SVM maintained its status as a robust model regardless of the minor improvement. But the Neural Network earned a greater KS statistic; it was not chosen as the champion model.

3.4.3 Decision Tree Validation

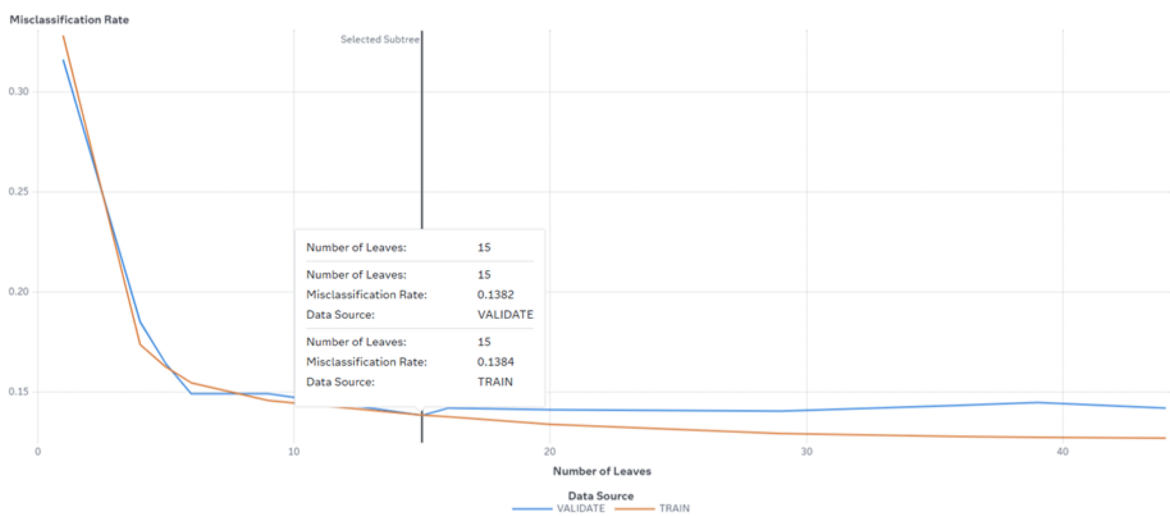


Figure 46: Misclassification Rate Graph (Tuned)

In the case of the decision tree having 15 leaves, there is no visible evidence of the presence of overfitting or underfitting in the results. Underfitting cannot be considered because of the relatively small misclassification rate in both training and validation samples. Overfitting should also be ruled out based on the near-equivalent errors in the training and validation datasets (0.1384 vs. 0.1382).

3.4.3.1 Fit Statistics

Fit Statistics

Target ...	Data Role	Partitio...	Formatt...	Number...	Average...	Divisor ...	Root Av...	Misclassification Rate
DropoutRisk	TEST	2	2	426	0.0885	426	0.2975	0.1056
DropoutRisk	TRAIN	1	1	2,609	0.1095	2,609	0.3310	0.1384
DropoutRisk	VALIDATE	0	0	1,389	0.1112	1,389	0.3334	0.1382

Figure 47: Fit Statistics of tuned Decision tree

Misclassification Rate is the smallest on the TEST data (10.56%) and almost the same for the TRAIN (13.84%) and VALIDATE (13.82%) datasets, demonstrating the good generalization capability of the model without overfitting.

3.4.3.2 Event Classification

Cutoff	Cutoff Source	Target Name	Response	Event	Value	Training Frequ...	Validation Freq...	Test Frequency	Training Percen...	Validation Perc...	Test Percentage
0.5000	Default	DropoutRisk	CORRECT	Not_Dropout	True Positive	1,651	893	285	94.1814	94	95
0.5000	Default	DropoutRisk	INCORRECT	Not_Dropout	False Negative	102	57	15	5.8186	6	5
0.5000	Default	DropoutRisk	CORRECT	Dropout	True Negative	597	304	96	69.7430	69.2483	76.1905
0.5000	Default	DropoutRisk	INCORRECT	Dropout	False Positive	259	135	30	30.2570	30.7517	23.8095
0.5300	KS	DropoutRisk	CORRECT	Not_Dropout	True Positive	.	876	.	.	92.2105	.
0.5300	KS	DropoutRisk	INCORRECT	Not_Dropout	False Negative	.	74	.	.	7.7895	.
0.5300	KS	DropoutRisk	CORRECT	Dropout	True Negative	.	316	.	.	71.9818	.
0.5300	KS	DropoutRisk	INCORRECT	Dropout	False Positive	.	123	.	.	28.0182	.
0.6700	KS	DropoutRisk	CORRECT	Not_Dropout	True Positive	1,526	.	266	87.0508	.	88.6667
0.6700	KS	DropoutRisk	INCORRECT	Not_Dropout	False Negative	227	.	34	12.9492	.	11.3333
0.6700	KS	DropoutRisk	CORRECT	Dropout	True Negative	671	.	104	78.3879	.	82.5397
0.6700	KS	DropoutRisk	INCORRECT	Dropout	False Positive	185	.	22	21.6121	.	17.4603

Figure 48: Event Classification of the tuned Decision tree

The table above depicts the false negative, true negative, false positive, and true positive of both the training and validation datasets.

False Negative (FN): 102 dropout students in the train dataset are incorrectly predicted to be not dropout. On the other hand, in the validate dataset, 57 dropout students are incorrectly predicted to be not dropout.

True Negative (TN): 597 dropout students in the train dataset are accurately predicted. In addition, in the validate dataset, 304 dropout students are accurately predicted.

False Positive (FP): 259 not dropout students in the train dataset are inaccurately predicted to be dropout. In addition, 135 not dropout students in the validate dataset are inaccurately predicted to be dropout.

True Positive (TP): 1,651 not dropout students in the train dataset are accurately predicted. In addition, 893 not dropout students in the validate dataset are accurately predicted.

3.4.3.3 ROC Reports

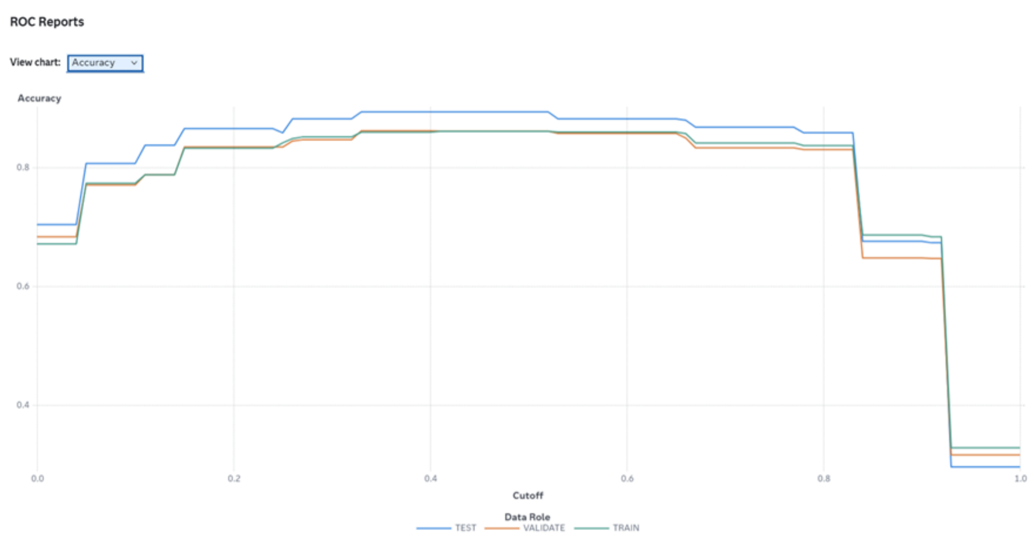


Figure 49: ROC Report showing Accuracy of the tuned Decision tree

This graph demonstrates changes in model Accuracy according to various Classification Cutoffs on Test, Validate, and Train datasets. This shows that the highest values of model accuracy are observed and remain almost unchanged at about 85% within the range of approximately 0.30 and 0.65 of the cutoff point. The extremely high correspondence in the results of all three data roles demonstrates good generalization of the model. However, using cutoffs close to 0.0 or over 0.85 would lead to extremely low model accuracy.

3.4.4 Forest Validation

Figure 50: Misclassification Rate Graph of tuned Forest model

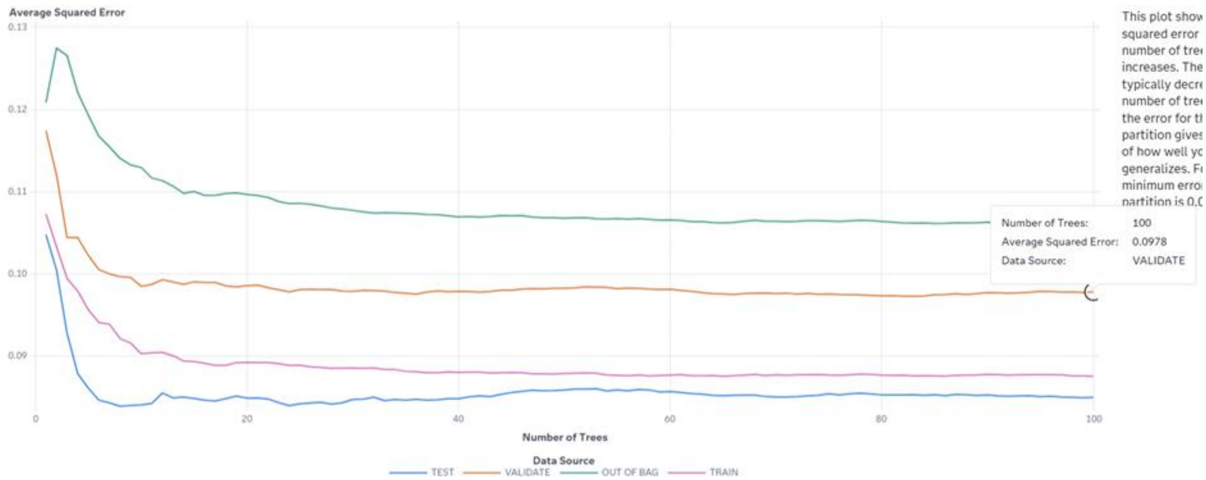


Figure 51: Average Square Error Graph of tuned Forest model

For all three datasets, Train, Validate, and Test, the error curves lie close to each other, showing no signs of separation even beyond the threshold of 100 trees. The high level of proximity and consistent parallel trends in error across the three data sets indicate that there is no overfitting or declining performance after this number of trees.

3.4.4.1 Fit Statistics

Data Role	Partitio...	Formatt...	Number...	Average...	Misclassification Rate
TEST	2	2	426	0.0850	0.1009
TRAIN	1	1	2,609	0.0876	0.1158
VALIDATE	0	0	1,389	0.0978	0.1253

Figure 52: Fit Statistics of tuned Forest model

The close alignment between the TRAIN (11.58%) and VALIDATE (12.53%) error rates confirms that the model generalizes well and is not overfitting.

3.4.4.2 Event Classification

Cutoff	Cutoff Source	Target Name	Response	Event	Value	Training Frequ...	Validation Freq...	Test Frequency	Training Percen...	Validation Perc...	Test Percentage
0.5000	Default	DropoutRisk	CORRECT	Not_Dropout	True Positive	1,680	894	289	95.8357	94.1053	96.3333
0.5000	Default	DropoutRisk	INCORRECT	Not_Dropout	False Negative	73	56	11	4.1643	5.8947	3.6667
0.5000	Default	DropoutRisk	CORRECT	Dropout	True Negative	627	321	94	73.2477	73.1207	74.6032
0.5000	Default	DropoutRisk	INCORRECT	Dropout	False Positive	229	118	32	26.7523	26.8793	25.3968
0.5900	KS	DropoutRisk	CORRECT	Not_Dropout	True Positive	.	861	.	.	90.6316	.
0.5900	KS	DropoutRisk	INCORRECT	Not_Dropout	False Negative	.	89	.	.	9.3684	.
0.5900	KS	DropoutRisk	CORRECT	Dropout	True Negative	.	345	.	.	78.5877	.
0.5900	KS	DropoutRisk	INCORRECT	Dropout	False Positive	.	94	.	.	21.4123	.
0.6400	KS	DropoutRisk	CORRECT	Not_Dropout	True Positive	.	.	269	.	.	89.6667
0.6400	KS	DropoutRisk	INCORRECT	Not_Dropout	False Negative	.	.	31	.	.	10.3333
0.6400	KS	DropoutRisk	CORRECT	Dropout	True Negative	.	.	107	.	.	84.9206
0.6400	KS	DropoutRisk	INCORRECT	Dropout	False Positive	.	.	19	.	.	15.0794
0.7100	KS	DropoutRisk	CORRECT	Not_Dropout	True Positive	1,542	.	.	87.9635	.	.
0.7100	KS	DropoutRisk	INCORRECT	Not_Dropout	False Negative	211	.	.	12.0365	.	.
0.7100	KS	DropoutRisk	CORRECT	Dropout	True Negative	736	.	.	85.9813	.	.
0.7100	KS	DropoutRisk	INCORRECT	Dropout	False Positive	120	.	.	14.0187	.	.

Figure 53: Event Classification of tuned Forest model

True Positive (TP): The model successfully predicts that the student does not have any risk of dropping out (Not_Dropout). It happens 1,680 times in the training phase and 894 times in the validation phase.

True Negative (TN): It refers to the situation where the model has made a right prediction regarding students who have already dropped out. True negatives take place 627 times during training and 321 times during validation.

False Positive (FP): It means that the model makes a mistake when predicting a dropout risk in students who do not really have any such risk. False positive takes place 229 times during training and 118 times during validation.

False Negative (FN): It takes place when the model fails to predict the drop-out risk in students who indeed have such risk, predicting their non-risky behavior instead. It happens 73 times during training and 56 times during validation.

3.4.4.3 ROC Reports

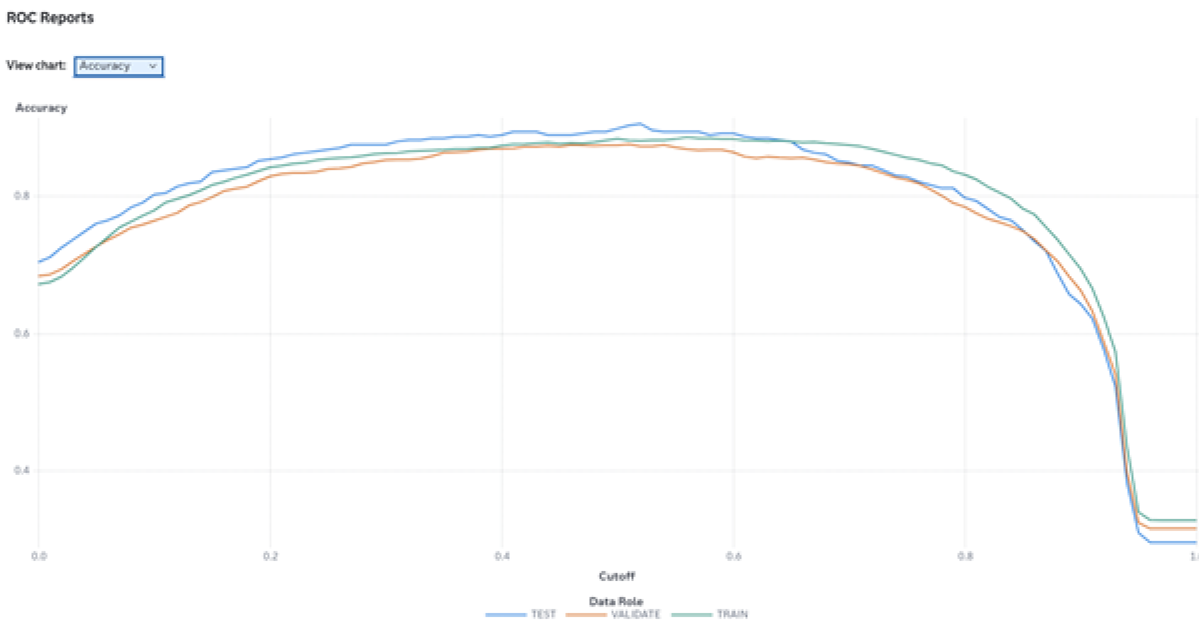


Figure 54: ROC Reports of tuned forest model showing Accuracy

All three data roles are close and overlap throughout the peak, which shows that the model has generalized well.

3.4.5 Logistic Regression Validation

Fit Statistics

Target ...	Data Role	Partitio...	Formatt...	Number...	Average...	Divisor ...	Root Av...
DropoutRisk	TEST	2	2	426	0.1088	426	0.3299
DropoutRisk	TRAIN	1	1	2,609	0.1204	2,609	0.3470
DropoutRisk	VALIDATE	0	0	1,389	0.1259	1,389	0.3548

Figure 55: Logistic Regression Fit Statistics

The Logistic Regression model was validated using the training, validation, and test datasets. The selected model used the Backward Elimination variable selection method because it produced the lowest validation misclassification rate among all tested configurations.

The training misclassification rate was 0.1033 (10.33%), while the validation misclassification rate was 0.1307 (13.07%). The test misclassification rate was 0.1188 (11.88%). The small difference of 2.74% between the training and validation misclassification rates indicates that the model generalizes well to unseen data and does not suffer from overfitting.

The validation accuracy of the selected Logistic Regression model was 86.93%, while the test accuracy was 88.12%. These results demonstrate that the model is stable and capable of effectively predicting student dropout risk.

Fit Statistics

Target ...	Data Role	Partitio...	Formatt...	Number...	Average...	Divisor ...	Root Av...	Misclas...	Multi-Cl...	K
DropoutRisk	TEST	2	2	426	0.0820	426	0.2863	0.1033	0.2778	
DropoutRisk	TRAIN	1	1	2,609	0.0993	2,609	0.3152	0.1307	0.3327	
DropoutRisk	VALIDATE	0	0	1,389	0.0941	1,389	0.3068	0.1188	0.3188	

Figure 56: Fit Statistics of the selected Logistic Regression model

The selected Logistic Regression model achieved a training misclassification rate of 10.33%, a validation misclassification rate of 13.07%, and a test misclassification rate of 11.88%. Since the validation error is only slightly higher than the training error, the model shows good generalization performance and does not exhibit significant overfitting.

Parameter Estimates

Effect	Parame...	t Value	Sign	Estimate	Absolut...	Standar...	Chi-Squ...
Curricular units 2nd sem (approv	Curricular units 2nd sem (approv	12.6643	+	0.5820	0.5820	0.0460	160.3837
Curricular units 2nd sem (enroll	Curricular units 2nd sem (enroll	11.0374	-	-0.4665	0.4665	0.0423	121.8242
Tuition fees up to date	Tuition fees up to date 0	9.4732	-	-1.9467	1.9467	0.2055	89.7408
Curricular units 1st sem (credit	Curricular units 1st sem (credit	6.3262	-	-0.2295	0.2295	0.0363	40.0209
Curricular units 1st sem (approv	Curricular units 1st sem (approv	6.0803	+	0.2909	0.2909	0.0478	36.9703
Intercept	Intercept	4.7027	+	1.4086	1.4086	0.2995	22.1151

Parameter Estimates

Effect	Parame...	t Value	Sign	Estimate	Absolut...	Standar...	Chi-Squ...
Curricular units 1st sem (approv	Curricular units 1st sem (approv	6.0803	+	0.2909	0.2909	0.0478	36.9703
Intercept	Intercept	4.7027	+	1.4086	1.4086	0.2995	22.1151
Age at enrollment	Age at enrollment	4.4854	-	-0.0373	0.0373	0.0083	20.1187
Mother's occupation	Mother's occupation	3.6616	+	0.0104	0.0104	0.0028	13.4077
Scholarship holder	Scholarship holder 0	3.2517	-	-0.5541	0.5541	0.1704	10.5739
Gender	Gender 0	2.1395	+	0.2717	0.2717	0.1270	4.5774
Tuition fees up to date	Tuition fees up to date 1	.	+	0	0	.	.
Scholarship holder	Scholarship holder 1	.	+	0	0	.	.
Gender	Gender 1	.	+	0	0	.	.

Figure 57: Parameter Estimates of the selected Logistic Regression model

The Parameter Estimates results show that academic performance variables are the strongest predictors of student dropout risk. Among the most influential variables are *Curricular Units 2nd Semester (Approved)*, *Curricular Units 2nd Semester (Enrolled)*, *Tuition Fees Up To Date*, and *Curricular Units 1st Semester (Approved)*. These variables have the highest statistical significance and contribute most to the model's predictions.

The results suggest that students with stronger academic performance and up-to-date tuition fee payments are less likely to be at risk of dropping out. In addition, factors such as Age at Enrollment, Mother's Occupation, Scholarship Holder status, and Gender were also retained in the final model, indicating that both academic and demographic characteristics influence student success.

Overall, the findings show that academic achievement is the most important factor in predicting student dropout risk, making Logistic Regression a useful and interpretable model for identifying students who may require early intervention.

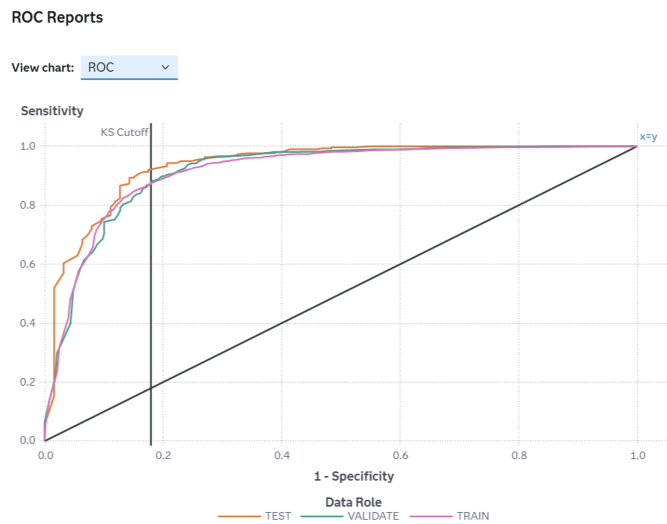


Figure 58: ROC Report of the selected Logistic Regression model

The ROC report demonstrates the classification performance of the Logistic Regression model across different probability cutoffs. The consistency of the results across the training, validation, and test datasets indicates that the model maintains stable predictive performance and generalizes well to unseen observations. The close alignment between the datasets further supports the conclusion that the model is neither overfitted nor underfitted.

3.4.6 Gradient Boosting Validation

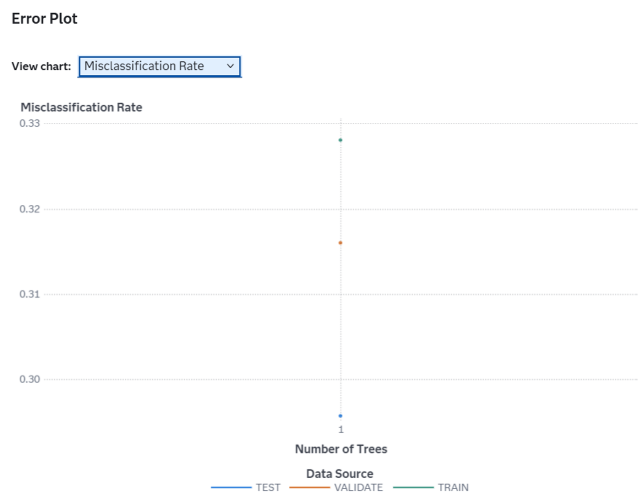


Figure 59: Gradient Boosting Misclassification Rate Graph

The Gradient Boosting model was validated using multiple parameter configurations. Among the evaluated models, the configuration with 50 trees, a learning rate of 0.1, and a maximum depth of 3 achieved the lowest validation misclassification rate and was therefore selected as the final model.

The selected model achieved a training misclassification rate of 0.1345 (13.45%), a validation misclassification rate of 0.1231 (12.31%), and a test misclassification rate of 0.0962 (9.62%). The close relationship between the training and validation errors indicates that the model generalizes effectively and does not suffer from overfitting.

Fit Statistics

Target ...	Data Role	Partitio...	Formatt...	Number...	Average...	Divisor ...	Root Av...	Misclas...	Multi-Cl...
DropoutRisk	TEST	2	2	426	0.2073	426	0.4553	0.2958	0.6050
DropoutRisk	TRAIN	1	1	2,609	0.2184	2,609	0.4673	0.3281	0.6282
DropoutRisk	VALIDATE	0	0	1,389	0.2143	1,389	0.4629	0.3161	0.6196

Figure 60: Fit Statistics of the selected Gradient Boosting model

The selected Gradient Boosting model produced a validation misclassification rate of 12.31%, corresponding to a validation accuracy of 87.69%. The difference between the training and validation misclassification rates is only 1.14%, demonstrating strong model stability and generalization capability.

Variable Importance

Variable Label	Role	Variable Name	Training Importance	Importance Standa...	Relative Importance
	INPUT	Curricular units 2nd sem (approv	248.5997	.	1
	INPUT	Tuition fees up to date	40.6191	.	0.1634
	INPUT	Course	18.6500	.	0.0750
	INPUT	Age at enrollment	3.1791	.	0.0128

Figure 61: Variable Importance of the selected Gradient Boosting model

The Variable Importance results show that *Curricular Units 2nd Semester (Approved)* is the most influential predictor in the Gradient Boosting model. Other important variables include *Tuition Fees Up To Date*, *Course*, and *Age at Enrollment*. These findings indicate that academic performance and financial status play a significant role in predicting student dropout risk.

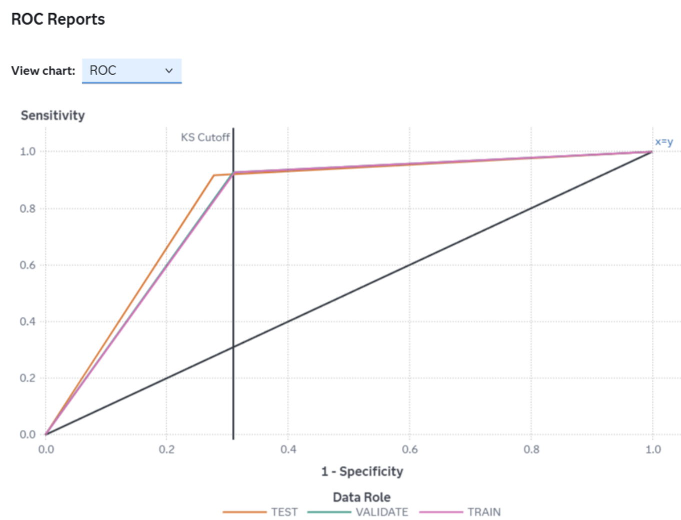


Figure 62: ROC Report of the selected Gradient Boosting model

The ROC report illustrates the classification performance of the Gradient Boosting model at different probability cutoffs. The consistency of the training, validation, and test results indicates that the model has learned meaningful patterns from the data while maintaining good generalization performance. The selected Gradient Boosting model achieved the best overall performance among the evaluated configurations and can therefore be considered a stable and well-fitted model for predicting student dropout risk.

3.5 Critical Interpretation of Outcomes

3.5.1 Mohib's Model Interpretation

Mohib designed and tested two predictive models which are Support Vector Machine (SVM) and Neural Network. The modified SVM generated a somewhat decreased validation misclassification rate of 0.1152 relative to the Neural Network's 0.1159.

Baseline Comparison Table:

Model	Validation Accuracy	Validation Misclassification	Validation AUC	Fit Status
Neural Network	88.41%	0.1159	0.9158	Good fit
SVM	88.34%	0.1166	0.9146	Good fit

The baseline layout with one hidden layer and fifty hidden nodes was preserved after a variety of neural network modifications were evaluated; however, none of them strengthened baseline model. The validation misclassification rate for the SVM had been enhanced from 0.1166 to 0.1152 by lowering the penalty parameter to 0.5.

Final Comparison of Mohib's Tuned Models:

Model	Final Setting	Validation Misclassification	Validation Accuracy	Decision
Neural Network	1 hidden layer, 50 hidden nodes	0.1159	88.41%	Strong model / Champion by KS
SVM	Linear kernel, penalty = 0.5	0.1152	88.48%	Best validation misclassification

The Neural Network was chosen as the project champion since it earned the greatest KS value on the testing split, regardless of the modified SVM having a somewhat reduced validation misclassification result. The two models did fairly well, but still depending on SAS Viya model comparison guidelines, the Neural Network delivered the most satisfactory outcome.

3.5.2 Zeba's Model Interpretation

Baseline Comparison Table:

Model	Validation Accuracy	Validation Misclassification	Fit Status

Decision Tree	B:2 N:12	85.8%	0.1418	Good fit
	B:6 N:17	85.1%	0.149	
Forest		86.5%	0.1346	Good fit

Final Comparison of Zeba's Tuned Models:

Model	**Final Setting**	**Validation Misclassification**	**Validation Accuracy**	**Decision**
Decision Tree	Branches- 4, Number of Leaves- 15	0.1382	86.2%	Strong Model
Forest	Trees- 100, Max. Depth- 10, Max. Branches- 2	0.1253	87.5%	Strong Model

3.5.3 Danish's Model Interpretation

Danish developed and evaluated two predictive models, namely Logistic Regression and Gradient Boosting. Both models achieved strong predictive performance in identifying students who are at risk of dropping out. The Gradient Boosting model demonstrated slightly better performance than Logistic Regression, producing a lower validation misclassification rate and higher validation accuracy.

Baseline Comparison Table:

Model	Validation Accuracy	Validation Misclassification	Fit Status
Logistic Regression (Stepwise)	83.86%	0.1614	Good Fit
Gradient Boosting (Default)	87.47%	0.1253	Good Fit

The baseline Logistic Regression model used the Stepwise variable selection method and achieved a validation misclassification rate of 0.1614. Several selection methods were evaluated, including Forward Selection and Backward Elimination. Among these methods, Backward Elimination produced the best performance and was selected as the final Logistic Regression model.

For Gradient Boosting, several configurations were tested by modifying the number of trees, learning rate, and maximum depth. The configuration consisting of 50 trees, a learning rate of 0.1, and a maximum depth of 3 produced the lowest validation misclassification rate and was selected as the final Gradient Boosting model.

Final Comparison of Danish's Tuned Models:

Model	Final Setting	Validation Misclassification	Validation Accuracy	Decision
Logistic Regression	Backward Elimination	0.1307	86.93%	Strong Model
Gradient Boosting	50 Trees, Learning Rate = 0.1, Maximum Depth = 3	0.1231	87.69%	Best Danish Model

The tuned Logistic Regression model successfully reduced the validation misclassification rate from 0.1614 to 0.1307, representing a significant improvement in predictive performance. The final model achieved a validation accuracy of 86.93% and demonstrated good generalization capability, as the difference between the training and validation misclassification rates remained below the 10% threshold.

The tuned Gradient Boosting model achieved the best overall performance among the evaluated configurations. It obtained the lowest validation misclassification rate of 0.1231 and the highest validation accuracy of 87.69%. The difference between the training and validation misclassification rates was only 1.14%, indicating that the model generalized well to unseen data and did not exhibit overfitting.

When comparing the two final models, Gradient Boosting slightly outperformed Logistic Regression in both validation accuracy and validation misclassification rate. Therefore, the

Gradient Boosting model was selected as the best-performing model among Danish's assigned predictive models.

4.0 Model Comparison

4.1 Overall Model Comparison

Model Comparison

Cham...	Name	Algorithm ...	KS (Youden)	Accuracy	Misclassification Rate ...	Average Squared Error	Area Under ROC	Cumulative Lift
★	Mohib - Neural Network	Neural Network	0.7730	0.8967	0.1033	0.0808	0.9358	1.4333
	Mohib - SVM	SVM	0.7492	0.8991	0.1009	0.1426	0.9372	1.4000
	Zeba - Decision Tree	Decision Tree	0.6690	0.8826	0.1174	0.0960	0.9058	1.4095
	Zeba - Forest	Forest	0.7325	0.8873	0.1127	0.0914	0.9268	1.4000
	Danish - Logistic Regression	Logistic Regression	0.6583	0.8521	0.1479	0.1088	0.8996	1.4667
	Danish - Gradient	Gradient Boosting	0.7508	0.8991	0.1009	0.0801	0.9327	1.4000

Figure 63: Overall Model Comparison Results for All Six Models

With the same generated dataset and testing split, the overall model comparison was conducted to compare all six models. Decision Tree, Neural Network, Forest, SVM, Logistic Regression, and Gradient Boosting were the models which were assessed. As shown in the comparison table, the models with the greatest efficiency and least misclassification rates include SVM, Neural Network, and Gradient Boosting.

89.67% accuracy tests and 0.1033 testing misclassification rate were obtained by the Neural Network model. The Gradient Boosting and SVM model both showed superior results, with testing misclassification rates of 0.1009 and testing accuracy of 89.91%.

Properties

Property Name	Property Value
selectionCriteriaClass	Kolmogorov-Smirnov statistic (KS)
selectionCriteriaInterval	Average squared error
selectionTable	Test
selectionDepth	10
cutoff	0.50

Figure 64: Model Comparison Selection Criteria

According to the Kolmogorov-Smirnov statistic (KS) for the assessment partition, SAS Viya chose the champion model, following the model comparison selection criteria. The cutoff value was selected as 0.50. The model with the greatest KS value was designated as the project champion model considering KS was the selection criterion.

4.2 Champion Model Selection

Project Summary

The champion model for this project is Mohib - Neural Network from the "Pipeline 1" pipeline. The model was chosen based on the KS (Youden) for the Test partition (0.77). 89.67% of the Test partition was correctly classified using the Mohib - Neural Network model. The five most important factors are Curricular units 2nd sem (approv, Curricular units 1st sem (approv, Tuition fees up to date, Age at enrollment, and Application mode.

Project Target:	DropoutRisk	**Project Champion:**	Mohib - Neural Network
Event Percentage:	67.8797%	**Created By:**	tp085584@mail.apu.edu.my
Pipelines:	1	**Modified:**	May 31, 2026, 07:36:18 AM

Figure 65: Project Champion Model Summary

The Mohib-Neural Network was picked by SAS as the champion model of the project. As it earned the maximum KS value on the test split, it was chosen. The model scored well in predicting the possibility of student dropout, accurately identifying 89.67% of the test data.

Since the model comparison criterion relied on the KS statistic, Neural Network was the preferred option even when SVM and Gradient Boosting had comparatively lower testing misclassification rates.

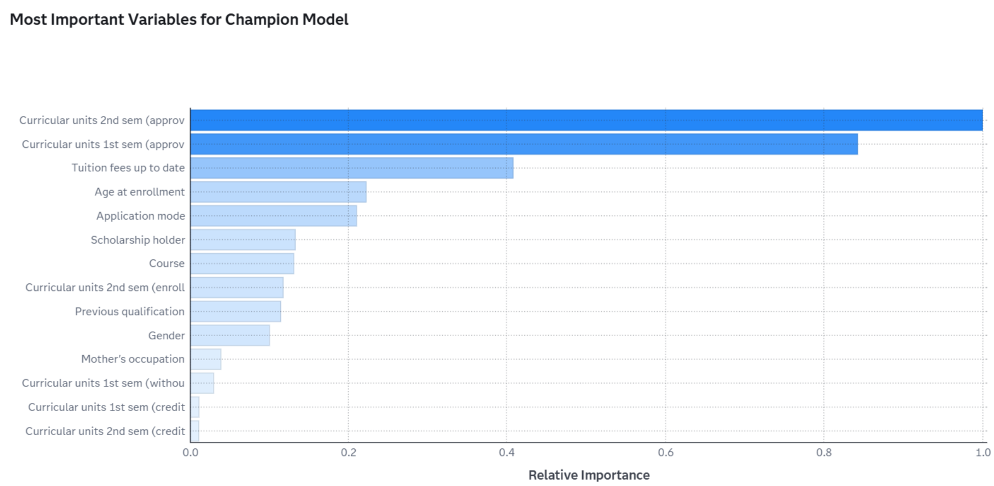


Figure 66: Important Variables for the Champion Model

The champion Neural Network model's key variables are presented in the important variables chart. Curricular units approved for the second term were by far the most substantial variable, next to Curricular unit's 1st Sem approved, Tuition fees up to date, Age at enrollment, and Application mode.

This indicates that wealth and academic success were key indicators of dropout risk. Financial struggles or poorer educational achievement can raise a student's probability of dropping out.

4.3 Discussion of Model Performance

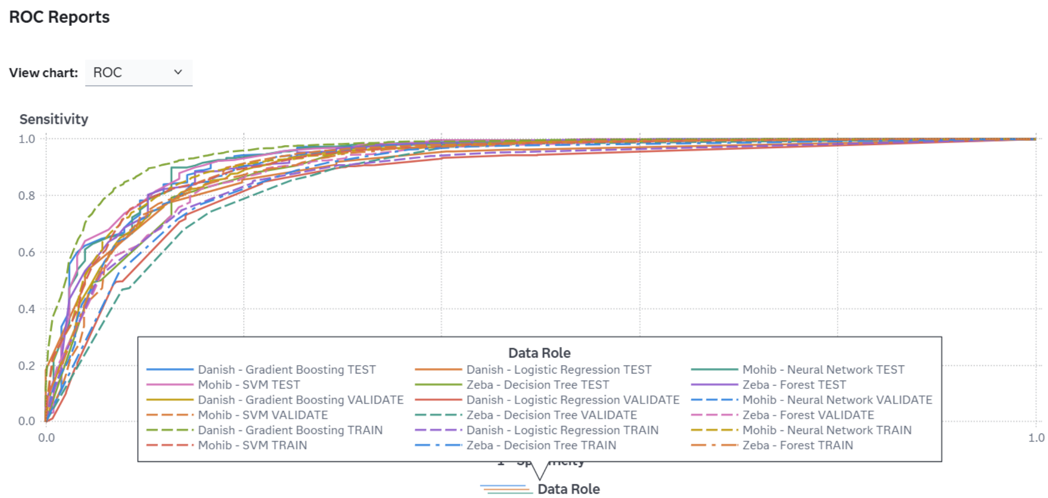


Figure 67: ROC Comparison of All Models

The ROC comparison found that most models performed good as their curves are stronger than the diagonal random line. This indicates that the models effectively divided Dropout and Non_Dropout students.

Therefore, Neural Network, SVM, and Gradient Boosting had the greatest outcomes. Neural Network was selected as the champion because of its high KS value, nevertheless SVM and Gradient Boosting also provided competitive outcomes with low testing misclassification rates. As an outcome, the final comparison illustrates that the models were suitable for estimating student dropout risk.

5.0 Conclusion

In this project, various predictive modelling methods such as Decision Tree, Forest, Gradient Boosting, Logistic Regression, Support Vector Machine (SVM) and Neural Network are used to predict the risk of dropping out from school with the data set STUDENT_DROPOUT_FINAL in the SAS Viya for Learners environment. All models were evaluated based on the performance measures – accuracy, misclassification rate, ROC values for the effectiveness of the models in predicting the class Dropout or Not_Dropout.

All the models were found capable of predicting patterns with respect to Student Attrition while not all the models were equally good as predictors of Student Attrition. The interpretable results were obtained using Decision Tree, Logistic Regression, and the prediction power was enhanced by using the ensemble learning Forest and Gradient Boosting. By observing the performance of all the models tested, the model trained by the Neural Network shows the best performance and is the Champion model. That's an indication of its ability to uncover the complex nature of the relationships between the data, and to make more precise predictions.

In summary, predictive analytics can help schools find and collect predictive data that would indicate a student is at risk and take the necessary steps.

6.0 Workload Matrix

Member	TP Number	Role	Signature
Mohib Bin Aamir	TP085584	Neural Network and Support Vector Machine models	Mohib
Zeba Waheed Parkar	TP084954	Decision Tree and Forest models	Zeba
Muhammad Danish Alfahmi Rusdi	TP077620	Logistic Regression and Gradient Boosting models	Danish

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